

A few-shot learning approach for database-free vision-based monitoring on construction sites

Jinwoo Kim^{a,c}, Seokho Chi^{b,c,*}

^a Department of Civil and Environmental Engineering, University of Michigan, Ann Arbor, MI 48109, USA

^b Department of Civil and Environmental Engineering, Seoul National University, Seoul 08826, Republic of Korea

^c Institute of Construction and Environmental Engineering, Seoul National University, Seoul 08826, Republic of Korea

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ABSTRACT

This paper proposes a few-shot learning approach that can successfully learn and detect new construction objects when only a few training data are given. The proposed approach includes few-shot model design and meta-learning processes. To validate the approach, the authors conducted experiments using a popular construction benchmark dataset, AIMDataset. Even if only 20 training images were provided to a new construction object, the few-shot learning could build an object detection model with the mean Average Precision of 73.1% on average, whereas the performance of the existing supervised learning was limited to 36.5%. The results imply that the proposed approach can successfully learn and detect new types of construction objects only with few labeled images given, enabling to reduce the number of training images while maximizing the model performance. It would be then possible to save human efforts required for data labeling and enhance the practicality of vision-based construction monitoring systems.

1. Introduction

Construction site monitoring, which is a process of measuring, analyzing, and improving project performance such as operational productivity and safety, is an important task for successful project management [1]. Project managers can evaluate the operational efficiency of input resources (e.g., direct work rate, hourly productivity) and discover root-causes of productivity loss [2–4]. For instance, if there are too many loaders that wait for trucks to arrive, they can make decisions to allocate more dump trucks on the site and minimize loaders' waiting times. Through continuous monitoring, it is also possible to identify potential hazardous objects (e.g., holes on worksites) and recognize unsafe behaviors of workers (e.g., working at heights without wearing personal-protective-equipment) [5]. Based on such on-site information, project managers can proactively remove hazardous objects and perform safety training to minimize construction accident risks. Accordingly, timely and accurate monitoring of construction sites can provide project managers with the information required for project controls and successful completion.

Due to the importance, project managers have directly visited construction sites and collected on-site information manually [6]. However, they faced difficulties in monitoring dynamic and large-scale

construction sites owing to time and cost limitations. For these reasons, they have installed video cameras at site, observed the collected images, and conducted vision-based monitoring remotely [7,8]. The remote vision-based monitoring has received more attention in practice [9,10] as the Korean Government has allowed construction firms to include the camera-installation costs in their safety management budget since 2016 [11]. Nonetheless, because project managers are required to manually observe and process a large amount of site videos to obtain on-site information (e.g., locations of input resources), the time and cost problems still arise. Hence, many researchers have developed numerous vision-based algorithms for automated video analysis and jobsite monitoring, enabling to capture on-site information from visual data without human intervention. The previous research efforts have demonstrated the high potential of vision-based algorithms for diverse monitoring purposes, such as productivity evaluation [12,13], safety analysis [14–16], and progress measurement [17].

However, as most state-of-the-art technologies rely on traditional deep learning algorithms, they require a significant amount of training data with accurate label information (e.g., object types and locations) to develop vision-based algorithms with high performance. It is thus essential to collect many construction images and annotate target objects (e.g., heavy equipment) and/or their operational information (e.g.,

* Corresponding author at: Department of Civil and Environmental Engineering, Seoul National University, Seoul 08826, Republic of Korea.

E-mail addresses: jinwooki@umich.edu (J. Kim), shchi@snu.ac.kr (S. Chi).

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object types and locations) on every single image frame. Since such human-dependent processes are time-consuming, expensive, and labor-intensive, it is still challenging to build an extensive and high-quality training image database (DB) [18–21]. Considering the fact that various types of construction resources exist on the site and they often vary from phase to phase, this issue becomes more problematic. Every time a new type of resources appears, project managers should build and update a training DB. For these reasons, it is frequent that the performance of vision-based monitoring would decline in practice. To address such problems, this paper proposes a few-shot learning approach for DB-free vision-based monitoring on construction sites. Herein, DB-free refers to a concept where the purpose is to minimize amount of training data and cost of data labeling, while maintaining the monitoring performance. To this end, this research builds on a few-shot learning approach that can successfully learn and detect new types of construction objects (i.e., *novel classes*) even if few training data are given, i.e., less than 30 images. Fig. 1 illustrates the concept of few-shot learning. The approach learns meta-knowledge (i.e., how to learn objects) from sufficient training data labeled with *base classes* (e.g., dump trucks, mixer trucks, rollers, and loaders), and leverages such knowledge to learn and detect *novel classes* (e.g., excavators) when a few training samples are given. In this paper, as shown in Fig. 1, the *base classes* are types of construction objects with sufficient labeled training data, while the *novel classes* mean types of objects with few training data.

This paper presents the following contributions. First, this study characterizes a novel technical approach that can learn new types of construction objects and achieve the promising performance of vision-based monitoring only with a few training images. Second, the approach can save time and costs for data labeling, and enhance the practical usability of vision-based systems in real-world construction sites. Third, to the best of authors' knowledge, this study is the first attempt to evaluate the feasibility of few-shot learning, which is one of the most appealing research topics in the field of deep learning and computer vision, to the construction domain. Particularly, the authors adapted and customized the few-shot learning to be applicable to construction environments, which have complex and different characteristics from others (e.g., indoor buildings, urban areas). Last, the quantitative results and theoretical findings of this study may play a crucial role as a baseline for future research directions in the area of vision-based construction monitoring. Future researchers can enhance their vision systems with the integration of the proposed few-shot learning.

2. Literature review

For automated construction monitoring, researchers have mainly focused on studying deep learning-based visual algorithms owing to their excellent performance. Kim et al. [22], for example, developed a region-based convolutional neural network (R-CNN) for detecting various types of construction equipment (e.g., excavators, dump trucks, mixer trucks, rollers, and loaders). Son et al. [23] investigated how to recognize construction workers under varying postures and viewpoints using the R-CNN. Fang et al. [24] further improved the previous R-CNN methods to classify construction workers and equipment simultaneously. You only look once (YOLO) architecture, which is lighter and faster than the R-CNN model, has also received attention from construction researchers. Nath et al. [25] used YOLO architecture for detecting construction workers, and Luo et al. [26] proposed a YOLO-based model to track the detected workers continuously. Pi et al. [27] also tested the applicability of YOLO model to the identification of diverse types of construction and building elements from aerial top-view images. The results of earlier works showed the high potential of the CNN models in complex construction environments and their robustness to the leading causes of the performance deviations of vision-based analysis (e.g., scale deviations, illumination effects).

The findings of existing studies led researchers to apply deep

learning-based visual algorithms to productivity and safety monitoring on construction sites. For automated work sampling and productivity analysis, Kim et al. [28] fed CNN-based equipment detection results in an earthmoving process simulation model to evaluate the operational efficiency. In a later study [29], they further applied the method for detecting dump truck's license plates and gathering their site-access log data. Luo et al. [30] developed a construction activity recognition method by integrating CNNs and relevance networks that analyze the types of construction objects (e.g., workers, equipment, and materials) and their spatial interrelations (e.g., size, proximity). Luo et al. [13] proposed a two-stream CNN approach to detect specific actions of construction workers from site surveillance videos, and they improved the method by appending Bayesian nonparametric learning to capture activities of the workers from far-field surveillance images [31]. As the CNN models often suffer from its inherent limitations, i.e., frame-by-frame time-independent analysis, some researchers made efforts to integrate time-series analysis models. Kim and Chi [32] combined CNN and double-layer long short term memory (LSTM) to incorporate sequential working patterns of earthmoving excavators (e.g., operation cycles), and Roberts and Golparvar-Fard [12] exploited a hidden Markov model to deal with temporal sequences of earthwork activities. Chen et al. [33] also designed a rule-based reasoning algorithm to interpret the time-series states of excavators from the CNN detection results. To understand the interactive operations of multiple construction resources, Cai et al. [34] developed a two-step LSTM method that categorizes working groups and their activity types. Bang and Kim [35] also integrated CNN and LSTM models to interpret object-to-object relationships (e.g., proximity) and extract their operational information (e.g., activity types) from site images.

Another mainstream of computer vision applications is construction safety monitoring. In one study [36], a CNN model was used to localize construction resources (e.g., workers, excavators, and wheel loaders) and measure their physical movements (e.g., proximities, speeds) for proactive responses to possible struck-by-object accidents. Son et al. [37] also analyzed the spatial collision risks between construction workers and equipment using CNN-based object detection results. As wearing personal-protective-equipment (PPE) is a major safety indicator, Fang et al. [15] utilized the CNN model to detect whether workers are wearing safety helmets or not. Wu et al. [38] further tested the performance of various CNN models on identifying workers' helmet-wearing states. In another study [14], they proposed a two-stream CNN approach to capture non-harness-wearing actions of construction workers for preventing falls from height. Similarly, Nath et al. [25] designed a two-step CNN architecture to recognize workers who do not wear safety helmet or vest, and Fang et al. [39] developed a vision-based method that detects diverse types of PPEs, e.g., helmets, harnesses, and anchorages, simultaneously. Other researchers also applied computer vision techniques to monitoring workers' unsafe postures and behaviors. Fang et al. [40] combined multiple CNN models for detecting non-certified workers who carry out a certain operation without permission (e.g., operating an excavator without driving license). Ding et al. [41] proposed a hybrid deep learning model composed of CNN and LSTM to recognize unsafe motions of workers while performing ladder-climbing tasks.

Existing studies have shown the exceptional performance of deep learning algorithms on vision-based construction monitoring. However, to generate a powerful deep learning model, it is inevitable to build a high quality and extensive training image DB. This process involves manually annotating target construction objects and/or their operational information, such as object types and locations, on every single image frame. Such manual processes not only require an excessive amount of time and effort but also have difficulty in incorporating a wide range of characteristics of different construction objects (e.g., different types, shapes, and colors of construction equipment). Thus, the practical use of vision-based monitoring on construction sites has been limited. To solve this problem, researchers have studied how to reduce

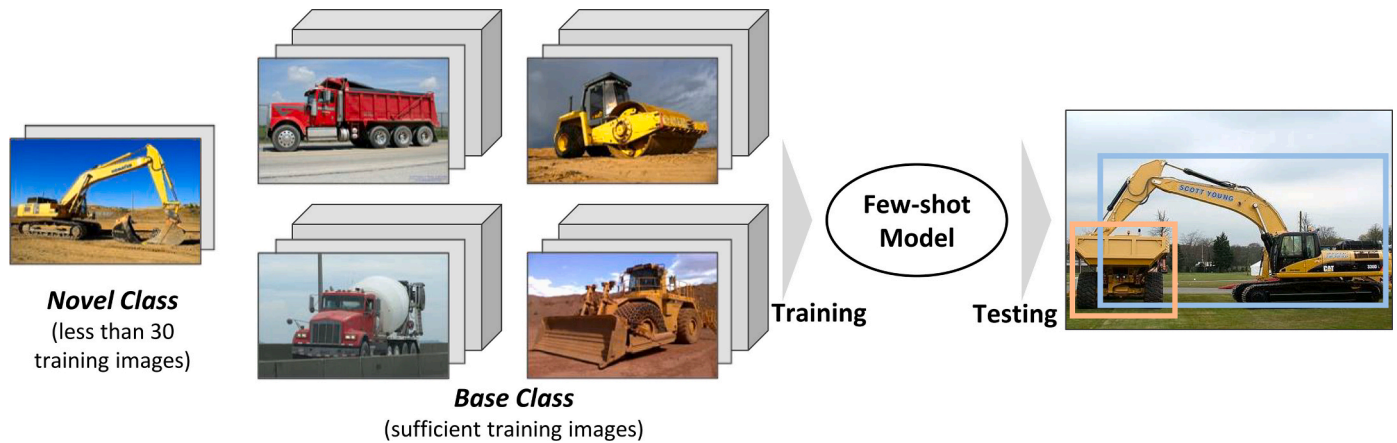


Fig. 1. Concept of few-shot learning.

the time and effort required to build a training DB. Liu and Golparvar-Fard [20], for instance, tested the practicality of crowdsourcing techniques, which are an effective way of outsourcing tedious image-annotation tasks to a crowd of non-expert individuals from an online community, such as Amazon Mechanical Turk. As they focused on annotating only workers and their action types, Wang et al. [19] improved the crowdsourcing technique to label complex safety-rule violations on construction images. However, as such crowdsourcing techniques still depend on human efforts and cannot reduce the required amount of training data, research efforts have been made to automate the image-annotation process. Soltani et al. [42] produced training data from a virtual equipment model and built a powerful object detector using the virtual images. Braun and Borrmann [43] also generated training images from building information modeling for detecting building elements, such as columns, walls, and slabs. Despite their encouraging proofs-of-concept, since real construction images have significantly different visual characteristics (e.g., textures and types of resources), it would not be guaranteed that deep learning models trained by virtual images have high performances. It may be also challenging to gather proper virtual models for every construction object and site. To minimize the amount of data labeling while maintaining the monitoring performance, this research builds upon a few-shot learning approach that can successfully learn and detect new types of construction objects (i.e., *novel classes*) even if few training samples are given, i.e., less than 30 images. Particularly, this study focuses on construction object detection, which is a crucial step for vision-based site monitoring.

3. Proposed approach: Few-shot learning

As shown in Fig. 1, the proposed approach aims to generate a few-shot model that can learn and detect new types of construction objects (i.e., *novel classes*) with only few images given. To achieve this, this research designs a few-shot learning model, and trains the model with the application of meta-learning which is the most critical algorithm for the few-shot learning. Details of model design and meta-learning are explained in the following sections.

3.1. Few-shot model design

This process designs a few-shot model that can learn meta-knowledge (i.e., how to learn target objects from image data) from training images labeled only with *base classes*, and leverage such knowledge to learn *novel classes*. The few-shot model consists of meta-feature extraction, class-attention, and prediction modules (see Fig. 2).

First, the meta-feature extraction module learns how to extract generalized visual features that can represent generic information about various construction objects; such generic features are called meta-

features that represent and explain general characteristics of the entire dataset (e.g., a set of object images) [45–47]. In this study, the module was built with the backbone of YOLOv2 (i.e., DarkNet-19) developed by Redmon Farhadi [48], which is one of the most popular and outstanding feature extraction architectures extensively used in both computer vision and construction domains [25–27]. Although the meta-features can provide generic visual information about all the target objects (both of *base* and *novel classes*), it may have difficulty in distinguishing different object classes each other due to a lack of class-specific cues. Such shortcoming can be counterbalanced by the class-attention module that captures class-specific information from the input images. Specifically, this module learns how to transform the input image to semantic class-attention vectors that assign greater weights on class-specific dimensions of the meta-feature vectors (Fig. 2). This allows to amplify visual signals that are more informative for a specific object class (e.g., excavator's unique characteristics such as its bucket and arm), thereby extracting class-specific visual cues from the meta-features and classifying different types of construction objects more effectively. The class-attention module, in this research, was designed with a light-weight CNN [44] and its initial configuration performed well in the authors' feasibility test to check computational efficiency and learning ease. Finally, the extracted meta-features and semantic class-attention vectors are channel-wise multiplied, and the multiplied channels are then fed into the prediction module to classify object types and localize their bounding boxes (Fig. 2). The authors designed the prediction module with two fully-connected layers of 128 hidden units and one softmax layer behind the last fully-connected layer. The model is learned by the meta-learning scheme as shown in Fig. 3 that is further explained in the next section.

Technically, the few-shot model predicts final outcomes from an input image as follows. Given the input image I_i for class $i = 1, 2, \dots, N$, the corresponding meta-feature F is generated by the meta-feature extraction module M (Eq. 1.1). Let us suppose that the support image I_i and their labeling information L_i (i.e., bounding box annotation) for class i are entered to the model; the support images mean a few labeled data given to *novel classes*. The class-attention module A receives one support image I_i and its corresponding label L_i as input, and embeds the image into a semantic class-attention vector w_i (Eq. 1.2); the dimension of each class-attention vector was carefully determined as 1024 based on the findings of the previous study [44] and the authors' heuristic experiments. Using the semantic class-attention coefficients w_i , it is possible to obtain the class-specific feature F_i for a *novel class* i (Eq. 1.3). Next, the feature vectors for all classes (i.e., a set of F_i for $i = 1, 2, \dots, N$) are fed into the prediction module P , which is a two-layer fully connected neural network with hidden units of 128. The detection results D_i , including the objectness score o_i (i.e., how likely the bounding box includes an object), bounding box coordinate (x_i, y_i, w_i, h_i) , and

classification confidence c_i , can be eventually obtained (Eqs. 1.4-1.5).

$$F = M(I_i) \quad (1.1)$$

$$w_i = A(I_i, L_i) \text{ for } \forall i \quad (1.2)$$

$$F_i = F \otimes w_i \text{ for } \forall i \quad (1.3)$$

$$D_i = (o_i, x_i, y_i, w_i, h_i, c_i) = P(F_i) \text{ for } \forall i \quad (1.4)$$

$$c_i = o_i \cdot \text{IoU} \cdot l_i \quad (1.5)$$

$$i = 1, 2, \dots, N \quad (1.6)$$

where $\otimes$ indicates channel-dimension wise multiplication, c_i indicates the softmax function-based probability of the corresponding object belonging to class i , IoU is the intersection over union between predicted boxes and ground truths, and l_i represents the conditional probability that the object belongs to class i , given an object is presence.

3.2. Meta-learning

The meta-learning, called “learning how to learn,” is the most critical algorithm of the few-shot learning; it guides the few-shot model to learn how to (1) extract meta-features and semantic class-attention vectors from input images using training data labeled only with *base classes* and (2) quickly adapt them for *novel classes* detection when a few support images are provided. Fig. 3 illustrates how the few-shot model, depicted in Fig. 2, is trained. The meta-learning is typically composed of two-phase learning steps: base learning and few-shot transfer learning.

First, the base learning plays an important role in learning how to extract meta-features and semantic class-attention vectors from input images, i.e., meta-knowledge. To do that, the base learning simulates multiple few-shot detection scenarios using sufficient training data labeled only with *base classes*. For example, in Fig. 3, a series of few-shot scenarios are simulated with the original *base classes* of “dump trucks,” “mixer trucks,” “rollers,” and “loaders.” The *base classes* are randomly split to *sub-base classes* and *sub-novel classes*, and each few-shot scenario can have k *sub-novel classes* and n support images per class, which is generally called k -way n -shot detection scenario. Hereinafter, during the base-learning phase, the *sub-base classes* mean types of construction objects with sufficient amount of training data, the *sub-novel classes* refer to types of objects with few training data. An example of 1-way 2-shot problems can be organized with the one *sub-novel class* of “dump

truck” given two labeled images, and the remaining three classes for “mixer trucks,” “rollers,” and “loaders” are assigned to the *sub-base classes*. In this case, the designed few-shot model (Fig. 2) is trained to detect “dump trucks” only with two labeled “dump truck” images by using the information learnt from “mixer truck,” “roller,” and “loader” *base classes*. This simulation allows the model to “learn how to learn” *novel classes*. By randomly selecting the values of k and n , the model can experience various few-shot scenarios and learn the meta-knowledge used for construction object detection. Based on the meta-knowledge, it would be possible to quickly adapt the designed model, as illustrated in Fig. 2, to the original *novel classes* (e.g., “excavators” in Fig. 3).

Once the base learning is completed, the few-shot transfer learning proceeds to finally learn a few support images for the *novel classes* and adapt the model to *novel class* detection scenarios. The training procedure is similar to the base learning, except that (a) the training data for *novel classes* (e.g., “excavators” in Fig. 3) are used, and (b) it takes considerably fewer iterations required for the model to converge due to a less amount of training data of the *novel classes*. It is also important to note that equal number of images for each object class are trained. In the case of k -way n -shot detection, n images are given to both *novel* and *base classes*.

For both training phases, the loss function for object detection is defined as a linear sum of the classification loss L_{cls} , the localization loss L_{loc} , and the objectness loss L_{obj} (Eq. 2.1). The classification loss can be computed with the cross-entropy loss over the calibrated softmax function score (Eq. 2.2), which is a widely-used loss function for a multi-classification problem. Eq. 2.3 explains how the localization loss, i.e., errors between the predicted bounding boxes and the ground truth, is calculated. The objectness loss is measured by Eq. 2.4.

$$L_{det} = L_{cls} + L_{loc} + L_{obj} \quad (2.1)$$

$$L_{cls} = - \sum_i \sum_j \mathbb{I}_{ij}^{obj} \log \left(\frac{e^{c_{ij}}}{\sum_n e^{c_n}} \right) \quad (2.2)$$

$$L_{loc} = \lambda_{loc} \sum_i \sum_j \mathbb{I}_{ij}^{obj} \left[\left(x_{ij} - \hat{x}_{ij} \right)^2 + \left(y_{ij} - \hat{y}_{ij} \right)^2 \right] + \lambda_{loc} \sum_i \sum_j \mathbb{I}_{ij}^{obj} \left[\left(\sqrt{w_{ij}} - \sqrt{\hat{w}_{ij}} \right)^2 + \left(\sqrt{h_{ij}} - \sqrt{\hat{h}_{ij}} \right)^2 \right] \quad (2.3)$$

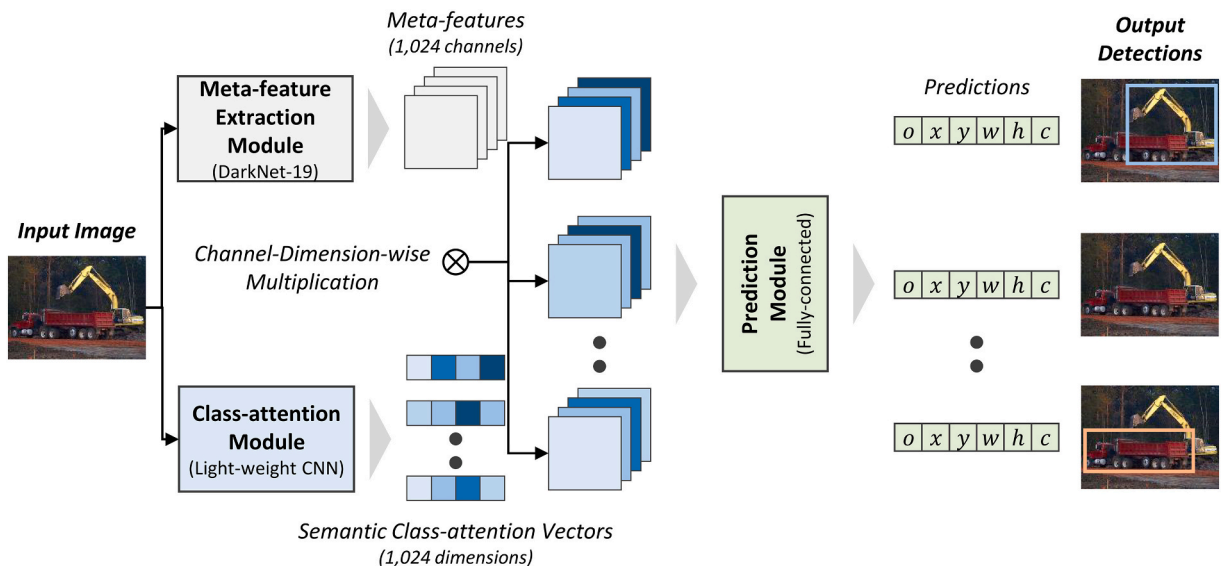


Fig. 2. The architecture of the proposed few-shot detection model (partially adopted from Kang et al. [44]).

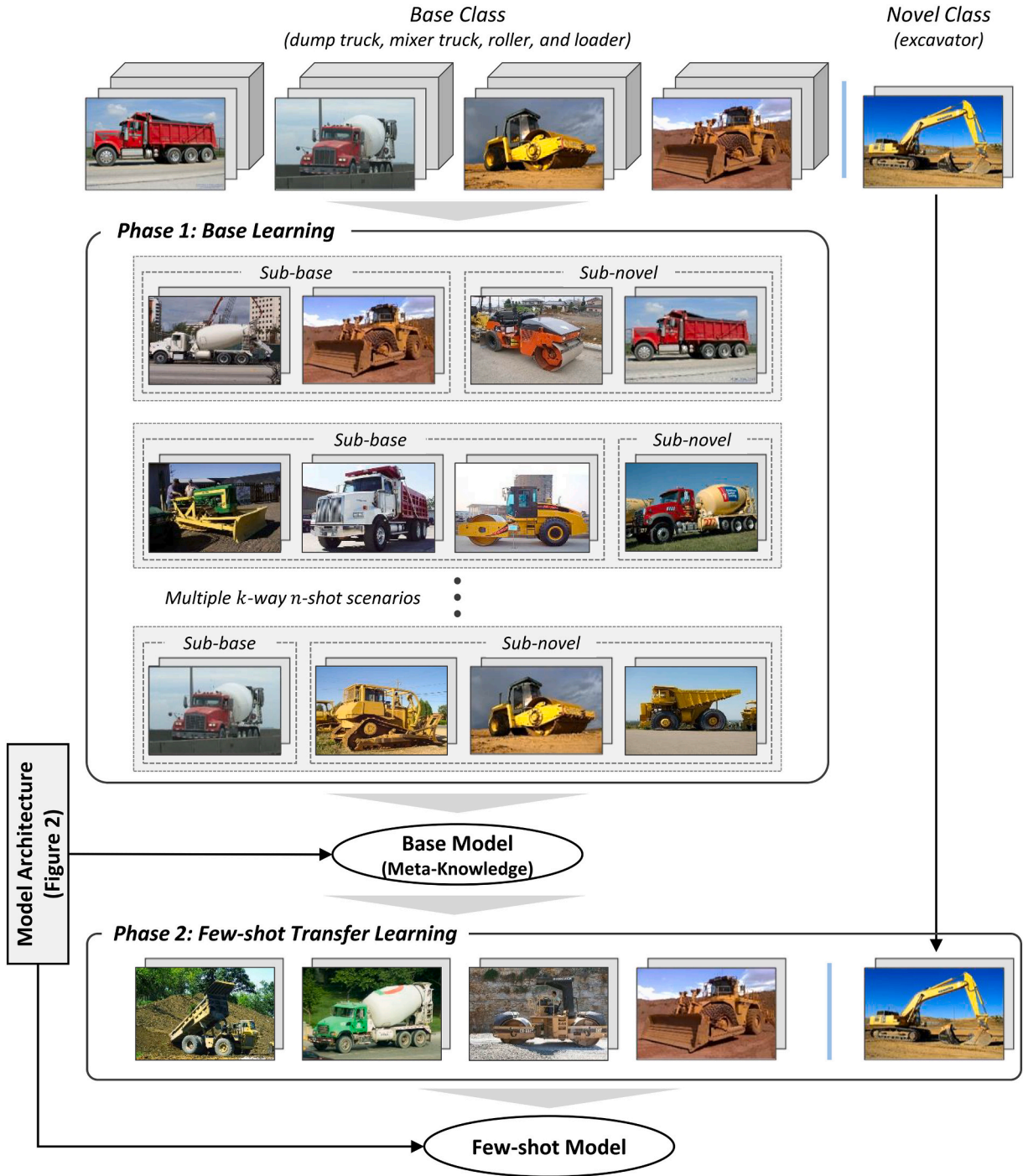


Fig. 3. Two-phase meta-learning scheme.

$$L_{obj} = \sum_i \sum_j \mathbb{I}_{ij}^{obj} \left(c_{ij} - \hat{c}_{ij} \right)^2 + \lambda_{no_obj} \sum_i \sum_j \mathbb{I}_{ij}^{no_obj} \left(c_{ij} - \hat{c}_{ij} \right)^2 \quad (2.4)$$

where the binary variables $\mathbb{I}_{ij}^{obj}$ and $\mathbb{I}_{ij}^{no_obj}$ represent whether the j -th boundary box in cell i is responsible for detecting the object or not, respectively; n is the number of object classes; c_{ij} indicates the confidence score of the box j in cell i ; x_{ij} , y_{ij} , w_{ij} , and h_{ij} denotes the center coordinates of the box j in cell i and its width and height; $\hat{x}_{ij}$ and $\hat{y}_{ij}$ are for the ground truth and the predicted box, respectively (likewise for y_{ij} , w_{ij} , h_{ij} , and c_{ij}); λ_{loc} and λ_{no_obj} mean the weighting parameter for the localization loss and non-objectness loss, respectively; here, cell i

indicates the i -th 7×7 two-dimensional grid in a given image, which is introduced in the original YOLOv2 research [48].

To minimize the defined loss function, the few-shot detection model is trained using stochastic gradient descent with a momentum of 0.9, a weight decay of 0.0005, and a batch size of 64. For base learning, a total of 80,000 iterations proceed with the various learning rates of 0.001 (~500 iterations), 0.0001 (~40,000 iterations), 0.00001 (~60,000 iterations), and 0.000001 (~80,000 iterations). The few-shot transfer learning is performed with a fixed learning rate of 0.001 and a total of 1500 iterations. These parameters worked well in the authors' heuristic experiments.

4. Experimental results and analysis

To validate the proposed approach, two different experiments were conducted: one with few-shot learning and the other without few-shot learning (i.e., with supervised learning). Supervised learning is a traditional approach of “learning target objects” appeared in a number of support images used for training a model, whereas few-shot learning focuses on “learning how to learn” even with a few support images provided. This research adopted YOLOv2, Faster R-CNN, and single shot detector (SSD) as the baseline detectors for supervised learning, because they are the most popular and outstanding architectures for visual object detection. Comparison with various deep architectures enabled the research team to evaluate the impacts of few-shot learning more comprehensively. When training object detectors using few-shot and supervised learning, the authors utilized the common data augmentation methods including image scaling, rotation, noising, translation, and horizontal/vertical flipping [49]. Thus, the authors can confirm both the applicability and practicality of the few-shot learning and evaluate its beneficial effects objectively.

For the experiments and result analyses, the research team used the AIMDataset that is one of the most popular construction benchmark datasets. The AIMDataset introduced by Kim et al. [22] contains five different classes of construction objects: “excavators,” “dump trucks,” “mixer trucks,” “rollers,” and “loaders.” In the dataset, the construction equipment has a range of working movements (e.g., loading, unloading) and viewpoints (e.g., front, front-left, rear, rear-right). As the images were collected from ImageNet [50] and there were various types of construction objects, the dataset would be enough to measure the generalization capability of the model [22]. The total number of samples is 2920, and widths and heights of the images range from 200 to 3651. In the authors’ experiments, the data were randomly divided into training data consisting of 2044 images (70%) and testing data consisting of 876 images (30%). Using the data, the research team trained and tested the models in diverse k -way n -shot scenarios, where the number of *novel classes* k was set from 1 to 3 and the number of support images n (given to each *novel class*) was set to 1, 2, 3, 5, 10, 20, and 30 considering the motivation of few-shot learning and the setup of previous studies; many researchers attempted to train and evaluate a deep learning model with less than 30 images [44,51,52]. The performances of the trained models were measured by the mean Average Precision (mAP), which is one of the most widely-used metrics in vision-based object detection challenges, e.g., PASCAL VOC [53]. Hereinafter, *base mAP* indicates the mAP for *base classes*, *novel mAP* implies the mAP for *novel classes*, and *all mAP* refers to the mAP for *all classes*.

4.1. Performance of the proposed approach

Table 1 provides experimental results according to different few-shot detection scenarios ($k = 1, 2, 3$ and $n = 1, 2, 3, 5, 10, 20, 30$). The developed model successfully detected construction equipment with the *all mAP* of 81.6% in the case of a 1-way 2-shot scenario. In detail, the *base mAP* recorded about 85.7%: “dump truck” (87.6%), “mixer truck” (90.8%), “roller” (78.5%), and “loader” (86.1%). The *novel mAP* was achieved to be 65.0% even if only two labeled images were given to the *novel class*, i.e., “excavators.” Furthermore, the *novel mAP* rapidly increased as only a few more images were added, and it eventually reached about 74.3% when 20 examples were trained. The proposed approach was also able to achieve 74.5% *novel mAP* and 80.0% *base mAP* (on the average for all $k = 1, 2, 3$) when more than 20 labeled data were used for each class. Despite the promising results, inevitable detection errors were also observed. Specifically, the model occasionally experienced difficulty in classifying dump trucks and mixer trucks due to their similar appearances. It might be also because 20 training images had a lack of detailed information to distinguish two similar objects. However, considering the fact that at least 300 image samples (per one object class) are generally required to obtain the detection performance over

75% [54–57], the proposed approach could reduce the number of training data to 1/15 (from 300 to 20 images), while achieving the comparable performance.

4.2. Performance with and without few-shot learning

The authors also observed significant positive effects of the proposed few-shot learning from the experiments. As presented in Fig. 4, the performances for *novel classes* were always higher for few-shot learning than existing supervised learning (i.e., YOLOv2, Faster R-CNN, and SSD) under the same experimental conditions (i.e., the same values of k and n when using the same data augmentation methods). Among the supervised learning models, Faster R-CNN had the lowest performance for all conditions, and SSD slightly worked better than YOLOv2. These results were consistent with the existing body of knowledge in computer vision and deep learning, i.e., complex models would not be trained well given few training data. The performance difference between the few-shot learning and the best supervised learning (i.e., SSD) was approximately 33.5% on the average, and these differences rose as the values of k and n increased; the peak value was nearly 42.5% at $k = 3$ and $n = 30$. This might be because the supervised learning method, which requires abundant labeled training data for all classes, faced difficulty in “learning *novel classes*” only with few images; however, the few-shot learning could “learn how to learn *novel classes*” using a few given samples. Owing to this ability, the proposed approach achieved the *novel mAP* over 78% at $n = 30$ (Table 1) even when the number of *novel classes* was set as 3 ($k = 3$), which is the case that more than the half of the total number of object classes were assigned to *novel classes* (the total number of object classes was five in the experiments). Reversely, the *novel mAP* of supervised learning remained in the vicinity of 30.2% under the same configuration. These results indicate that the few-shot learning approach not only reduces the number of training data required to generate a deep learning model but also quickly learn to detect *novel classes* even if only a few training data are given. With these benefits, it may be applicable to real-world construction environments where there are many novel and base objects simultaneously.

4.3. Generalizability and expandability of the proposed approach

To further confirm the generalizability and expandability of the proposed approach, the authors conducted an additional experiment using different image data collected from actual construction sites. A total of 5000 image frames were collected with 1280×720 resolution, and the collected data were randomly split into 3500 (70%) training and 1500 (30%) testing images. The image data included two different types of construction objects (i.e., “excavators,” “dump trucks”), and object detection models were trained in the same manner with the prior experiment. Table 2 shows the *novel mAPs* of few-shot and supervised learning approaches according to different n values; the number of novel classes was set as 1 (“excavators”). For all cases, the few-shot learning approach outperformed the supervised learning, and the performance difference was 32.5% on average. In line with the previous experimental results, the performance differences tended to rise as the value of n increased, and the peak value was nearly 42.2% at $n = 10$. The highest performance of the few-shot learning was discovered as 72.5%. These experimental results indicated that the proposed approach have a potential to be generalized and applied to image data collected from different monitoring environments.

5. Results and discussion

It was found that the proposed approach can learn the meta-knowledge (i.e., how to learn target objects) from training data labeled only with *base classes*, and leverage such knowledge to detect *novel classes* when a few support images are given. The trained model effectively localized the different types of construction objects, such as

“excavators,” “dump trucks,” “mixer trucks,” “rollers,” and “loaders,” with the *all mAP* of 80.4% in the 1-way 20-shot scenario (Table 1). Fig. 5 displays the detection results of the developed model. As can be seen, the model performed well even when the heavy equipment presents a range of working postures and visual characteristics (e.g., colors, shapes, sizes) under different camera positions and viewpoints. Specifically, although the dump trucks in the figure had different visual appearances, all of them were classified well by the model. The model was also able to detect construction objects in cluttered background and correctly classify the equipment that has similar visual characteristics (e.g., dump trucks-mixer trucks, loaders-rollers). In addition, at the example of the bottom-left of Fig. 5, the model succeeded in localizing novel (i.e., excavators) and base objects (i.e., dump trucks) in the same image. The fourth image in the first row also shows that the developed model could localize the excavator that was partially occluded by the loader. Similarly, the model detected the roller that was not fully visible from the image (the last image in the first row).

The results also demonstrated the substantial benefits of the proposed few-shot learning. As reported in the experiments using the AIMDataset, the *novel mAP*s were always higher when using few-shot learning than supervised learning (Fig. 4), which means that the proposed approach can build a more powerful object detection model given the equal number of training data. When k is equal to 1 (i.e., the most straightforward few-shot scenario), the *novel mAP* recorded approximately 40.0% only with one image sample ($n = 1$), and the *all mAP* was approximately 71.0%. The *all mAP* of supervised learning was also achieved to be 74.0%, but the *novel mAP* was limited to only 36.4%. It was notable that the performance disparity for *novel classes* had been more striking as more support images were added. When n ranges from 1 to 30, the performance difference between the two approaches was about 21.5% on the average, and the peak value was 31.6% at $n = 5$. Moreover, the performance disparity had become more significant, as the number of *novel classes* increased and the few-shot problem became more challenging. Indeed, when k was changed from 1 to 3, the *novel mAP* of the supervised learning approach dropped from 36.4% to 9.1% at $n = 1$, whereas the performance loss of few-shot learning was comparatively insignificant (40.0% $\rightarrow$ 27.7%). The performance of few-shot learning even recovered to 53.1% once the value of n became 2, and it finally peaked the *novel mAP* of 78.0% when 30 image samples were provided. In contrast, the *novel mAP* of supervised learning remained in a maximum of 29.5% although 30 images were used, and this eventually resulted in the performance difference of 38.8% on the average. These findings imply that the proposed approach can fully leverage labeled data for *base classes* to learn meta-knowledge, and successfully transfer the knowledge to detect *novel classes* with very few given examples, i.e., less than 30 images. It also seems that the few-shot learning performs well even in the situation that many *novel classes* (i.e., more than the half of total number of given object classes) need to be detected.

The aforementioned performance differences between the few-shot and supervised learning can be qualitatively explained with t-distributed Stochastic Neighbor Embedding (t-SNE) [58]. The t-SNE visualizes the semantic class-attention vectors for each object class, and it can be

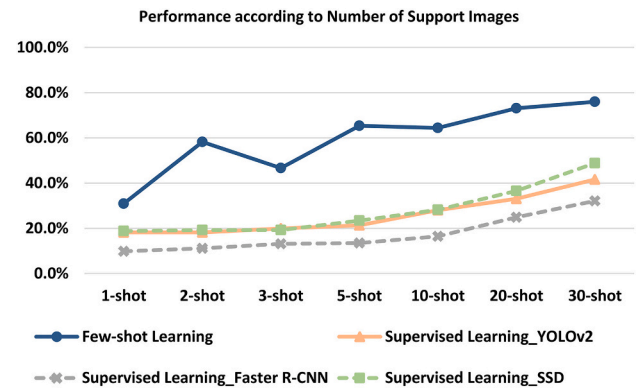


Fig. 4. Comparative analysis for the novel mAPs of few-shot and supervised learning.

used to interpret how much the vectors are well-trained for object classification. Fig. 6 illustrates the t-SNE of the 3-way 20-shot scenario. In the figure, the closer the data points for the same classes are and the greater the distances between different classes (e.g., “excavators” and “dump trucks”) become, it means that the trained model can classify target objects more effectively. In the case of few-shot learning, most of the data points tended to form clusters, and the intercluster distances were quite large (see Fig. 6(a)). It is easily noticeable that the *novel classes* of “excavators,” “dump trucks,” and “mixer trucks” are evenly distributed, which denotes that the model knows how to distinguish those objects. In contrast, as shown in Fig. 6(b), the supervised learning had quite large overlapping areas between different clusters. The clusters of *novel classes* (“excavators,” “dump trucks,” and “mixer trucks”) overlaid different object classes substantially. This indicates that the model might face difficulties in extracting class-specific visual features and classifying different object classes each other, resulting in frequent misclassifications and performance decrease.

6. Research contributions and future works

In this section, the authors discuss theoretical findings and contributions of this study, and research challenges and future works are also suggested in the perspective of practical applicability.

First, this study revealed that it is possible to build a promising deep learning model only using a few training data. These findings can bring a positive impact on implementing vision-based monitoring in real-world construction sites. To be specific, regardless of any new types of construction resources appeared during different construction phases, the proposed few-shot learning approach can quickly adapt to such dynamic conditions and monitor new objects for which a sufficient amount of training data are not necessary. This nature can allow site managers to reduce their efforts and time required for additional data collection and annotation, which would be very labor-intensive and time-consuming. Even for a particular object (e.g., excavators) having many training

Table 1
Experimental results of the proposed approach.

k -way n -shot scenarios		n : number of support images						
k : number of novel classes		1-shot	2-shot	3-shot	5-shot	10-shot	20-shot	30-shot
1-way	Novel mAP	40.0%	65.0%	54.7%	68.1%	65.4%	74.3%	74.5%
	Base mAP	78.7%	85.7%	80.3%	73.9%	77.9%	81.9%	81.4%
	All mAP	71.0%	81.6%	75.1%	72.7%	75.1%	80.4%	80.1%
2-way	Novel mAP	25.1%	56.6%	44.7%	63.2%	65.4%	73.1%	75.5%
	Base mAP	68.9%	85.3%	69.8%	74.0%	77.7%	80.5%	80.9%
	All mAP	51.4%	73.9%	59.7%	69.7%	72.3%	77.5%	78.8%
3-way	Novel mAP	27.7%	53.1%	40.6%	64.8%	62.5%	71.8%	78.0%
	Base mAP	61.6%	70.6%	65.7%	71.1%	72.8%	77.1%	78.1%
	All mAP	41.3%	60.1%	50.6%	67.3%	66.6%	73.9%	78.0%

Table 2

Experimental results of few-shot and supervised learning.

	1-shot	2-shot	3-shot	5-shot	10-shot	20-shot	30-shot
Few-shot Learning	35.2%	42.7%	49.9%	56.1%	64.7%	68.5%	72.5%
Supervised Learning_YOLOv2	13.2%	15.1%	16.8%	18.5%	22.5%	32.8%	43.2%

samples available, the proposed approach can also reduce human efforts required to annotate a target object and/or their operational information (e.g., object types and locations). These technical advantages would be beneficial when developing and updating a training DB phase-by-phase, and site managers can perform vision-based monitoring more efficiently. According to the authors' interviews with construction experts, it was reported that the DB development and update is one of the most crucial obstacles that limit the use of vision systems in their projects. In this regard, it is believed that the ability to reduce the required number of training images to 1/15 compared to previous methods would be very effective and helpful. Further, the theoretical findings of this study can provide valuable insights and future directions to researchers in the field of vision-based construction monitoring.

Next, when considering on-site applications of these findings, it should be carefully concerned that there was a technical limitation to train a very robust model. For instance, the research team faced difficulty in maximizing the model performance with the *mAP* over 90% only using few images. It was also reported that the proposed few-shot learning approach (*all mAP*: 87.8%) was not able to outperform the existing supervised learning (*all mAP*: 91.6%) when using all available training data in the AIMDataset (2044 image frames). It may be because the goal of existing supervised learning is to obtain the maximum performance by using a large amount of training data, whereas the few-shot learning focuses on training a comparable model given limited number of images. Particularly, the few-shot model occasionally faced difficulty in precisely localizing multiple objects (more than three) in crowded scene images; as can be seen in Fig. 7(a), the model could predict the types of construction objects (e.g., "dump trucks," "loaders") that exist in the image, but the localization was incorrect. One effective alternative for this localization issue is to adopt a domain adaptation technique. According to transfer learning theory [59] and the findings of existing empirical studies [22,25], object detection models (composed of two different modules of object localization and classification) can share

their localization networks with each other. This means that the localization network pretrained by a different domain dataset (e.g., ImageNet) can be used for construction object localization without or with little fine-tuning. By adopting such domain adaptation technique, it would be possible to address the localization problem more effectively.

It was also observed that the performance of few-shot learning was sensitive to the support images sampled and used in the stage of meta-learning. In the authors' experiments, the model performance dropped from 58.2% to 46.7% when the number of support images varied from 2 to 3 (Fig. 4). This observation was inconsistent with common sense that the more number of images are used, the more powerful model is learned. It might be because the support images (that were randomly sampled for meta-learning) did not provide rich and meaningful information about target objects. Fig. 7(b) shows that three support images, which were used in the authors' experiment ($n = 3$), included visual information only about rear side of dump trucks, and therefore the model could not learn their diverse characteristics. In this case, detection errors mainly occurred when dump trucks were captured from other viewpoints, such as front and/or side views. To address this issue, future studies can adopt an active learning scheme that can select the most meaningful-to-learn instances from a given training dataset. In the prior example, it would be more effective if examining the visual similarities of given images and selecting the three most representative and visually distinct instances (e.g., left-hand side, front-view, rear side). One recent study also reported that an active learning scheme is effective in training a deep learning model more stably and powerfully [60].

7. Conclusions

This paper presented a few-shot learning approach to achieve DB-free vision-based monitoring on construction sites. The approach includes few-shot model design and meta-learning processes. As observed in the experiments, the few-shot model was able to learn the meta-

**Fig. 5.** Examples of the experimental results in the construction object detection.

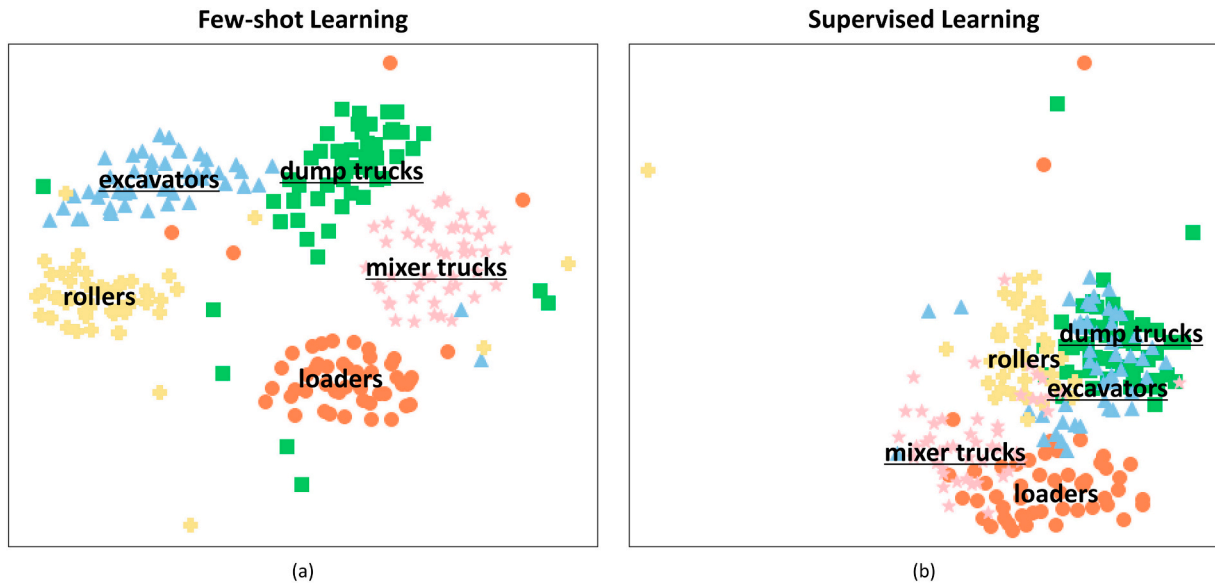


Fig. 6. t-SNE plot of the semantic class-attention vectors: (a) Few-shot learning; (b) Supervised learning.



Fig. 7. Errors examples: (a) localization errors; (b) sampling bias.

knowledge from training data labeled only with *base classes*, and the meta-learning enabled to successfully transfer such knowledge to detect *novel classes* when a few images are given. In the 3-way 30-shot scenario, the few-shot detector localized the *novel classes* with the *novel mAP* over 78.0%, while the performance of supervised learning was limited to only 29.5%. It means that the proposed approach can train a much more powerful object detector than the existing method, given the same number of training data. This performance difference was also qualitatively explained with the t-SNE visualization (Fig. 6). The semantic class-attention vectors of few-shot learning were evenly distributed and helped to classify various types of objects, but those of supervised learning had large overlapping areas and thus classification errors occurred frequently. With the benefits of the proposed approach, this study made the following contributions. First, this research introduced a novel technical framework that can significantly reduce the required number of training images while maintaining the performance of vision-based monitoring. Second, the framework can save time and costs required for data labeling, thereby enhancing the practical acceptability of vision systems on construction sites. Third, to the best of authors' knowledge, this was the first attempt to apply few-shot learning, which is one of the most attracting and emerging pattern recognition algorithms, in the construction domain. The authors customized the few-shot learning to be applicable to construction environments, and provided quantitative results to demonstrate its technical feasibility. Last, the findings of this study can build the foundation for future research in the area of not only vision-based construction monitoring but also other research fields, e.g., signal processing [61–63] and text-mining [64–66].

For developing a deep learning model, future studies can combine it with the few-shot learning when only few training data are available.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] J. Kim, Automated Construction Site Monitoring: Multi Vision-Based Operational Context Analysis for Enhancing Earthmoving Productivity, Seoul National University, 2019. <http://hdl.handle.net/10371/161874>. accessed May 14, 2020.
- [2] J. Yang, M.W. Park, P.A. Vela, M. Golparvar-Fard, Construction performance monitoring via still images, time-lapse photos, and video streams: now, tomorrow,

- and the future, *Adv. Eng. Inform.* 29 (2015) 211–224, <https://doi.org/10.1016/j.aei.2015.01.011>.
- [3] E. Rezazadeh Azar, Semantic annotation of videos from equipment-intensive construction operations by shot recognition and probabilistic reasoning, *J. Comput. Civ. Eng.* 31 (2017), 04017042, [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000693](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000693).
 - [4] D. Roberts, W.T. Calderon, S. Tang, M. Golparvar-fard, Vision-based construction worker activity analysis informed by body posture, *J. Comput. Civ. Eng.* 34 (2020), 04020017, [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000898](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000898).
 - [5] J. Seo, S. Han, S. Lee, H. Kim, Computer vision techniques for construction safety and health monitoring, *Adv. Eng. Inform.* 29 (2015) 239–251, <https://doi.org/10.1016/j.aei.2015.02.001>.
 - [6] Y. Ham, K.K. Han, J.J. Lin, M. Golparvar-Fard, Visual monitoring of civil infrastructure systems via camera-equipped unmanned aerial vehicles (UAVs): a review of related works, *Visual. Eng.* 4 (2016) 1–8, <https://doi.org/10.1186/s40327-015-0029-z>.
 - [7] J. Kim, Y. Ham, Y. Chung, S. Chi, Camera placement optimization for vision-based monitoring on construction sites, in: 2018 Proceedings of the 35th International Symposium on Automation and Robotics in Construction, International Association for Automation and Robotics in Construction, Berlin, Germany, 2018, pp. 748–752, <https://doi.org/10.22260/ISARC2018/0102>.
 - [8] J. Kim, Y. Ham, Y. Chung, S. Chi, Systematic camera placement framework for operation-level visual monitoring on construction jobsites, *J. Constr. Eng. Manag.* 145 (2019), 04019019, [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0001636](https://doi.org/10.1061/(ASCE)CO.1943-7862.0001636).
 - [9] J. Gong, C.H. Caldas, Computer vision-based video interpretation model for automated productivity analysis of construction operations, *J. Comput. Civ. Eng.* 24 (2010) 252–263, [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000027](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000027).
 - [10] J. Yang, P. Vela, J. Teizer, Z. Shi, Vision-based tower crane tracking for understanding construction activity, *J. Comput. Civ. Eng.* 28 (2014) 103–112, [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000242](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000242).
 - [11] Korea Construction Technology Promotion Act, Enforcement decree article 98 and 99, statutes of the Republic of Korea. <http://law.go.kr/법령/건설기술관리법>, 2016 (accessed January 28, 2019).
 - [12] D. Roberts, M. Golparvar-Fard, End-to-end vision-based detection, tracking and activity analysis of earthmoving equipment filmed at ground level, *Autom. Constr.* 105 (2019) 102811, <https://doi.org/10.1016/j.autcon.2019.04.006>.
 - [13] X. Luo, H. Li, D. Cao, Y. Yu, X. Yang, T. Huang, Towards efficient and objective work sampling: recognizing workers' activities in site surveillance videos with two-stream convolutional networks, *Autom. Constr.* 94 (2018) 360–370, <https://doi.org/10.1016/j.autcon.2018.07.011>.
 - [14] W. Fang, L. Ding, H. Luo, P.E.D. Love, Falls from heights: a computer vision-based approach for safety harness detection, *Autom. Constr.* 91 (2018) 53–61, <https://doi.org/10.1016/j.autcon.2018.02.018>.
 - [15] Q. Fang, H. Li, X. Luo, L. Ding, H. Luo, T.M. Rose, W. An, Detecting non-hardhat-use by a deep learning method from far-field surveillance videos, *Autom. Constr.* 85 (2018) 1–9, <https://doi.org/10.1016/j.autcon.2017.09.018>.
 - [16] M.-W. Park, N. Elsafty, Z. Zhu, Hardhat-wearing detection for enhancing on-site safety of construction workers, *J. Constr. Eng. Manag.* 141 (2015), 04015024, [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0000974](https://doi.org/10.1061/(ASCE)CO.1943-7862.0000974).
 - [17] A. Dimitrov, M. Golparvar-Fard, Vision-based material recognition for automated monitoring of construction progress and generating building information modeling from unordered site image collections, *Adv. Eng. Inform.* 28 (2014) 37–49, <https://doi.org/10.1016/j.aei.2013.11.002>.
 - [18] J. Kim, S. Chi, Adaptive detector and tracker on construction sites using functional integration and online learning, *J. Comput. Civ. Eng.* 31 (2017), 04017026, [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000677](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000677).
 - [19] Y. Wang, P.C. Liao, C. Zhang, Y. Ren, X. Sun, P. Tang, Crowdsourced reliable labeling of safety-rule violations on images of complex construction scenes for advanced vision-based workplace safety, *Adv. Eng. Inform.* 42 (2019) 101001, <https://doi.org/10.1016/j.aei.2019.101001>.
 - [20] K. Liu, M. Golparvar-Fard, Crowdsourcing construction activity analysis from jobsite video streams, *J. Constr. Eng. Manag.* 141 (2015), 04015035, [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0001010](https://doi.org/10.1061/(ASCE)CO.1943-7862.0001010).
 - [21] J. Kim, Visual analytics for operation-level construction monitoring and documentation: state-of-the-art technologies, research challenges, and future directions, *Front. Built. Environ* 6 (2020) 575738, <https://doi.org/10.3398/fbuil.2020.575738>.
 - [22] H. Kim, H. Kim, Y.W. Hong, H. Byun, Detecting construction equipment using a region-based fully convolutional network and transfer learning, *J. Comput. Civ. Eng.* 32 (2018), 04017082, [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000731](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000731).
 - [23] H. Son, H. Choi, H. Seong, C. Kim, Detection of construction workers under varying poses and changing background in image sequences via very deep residual networks, *Autom. Constr.* 99 (2019) 27–38, <https://doi.org/10.1016/j.autcon.2018.11.033>.
 - [24] W. Fang, L. Ding, B. Zhong, P.E.D. Love, H. Luo, Automated detection of workers and heavy equipment on construction sites: a convolutional neural network approach, *Adv. Eng. Inform.* 37 (2018) 139–149, <https://doi.org/10.1016/j.aei.2018.05.003>.
 - [25] N.D. Nath, A.H. Behzadan, S.G. Paal, Deep learning for site safety: real-time detection of personal protective equipment, *Autom. Constr.* 112 (2020) 103085, <https://doi.org/10.1016/j.autcon.2020.103085>.
 - [26] X. Luo, H. Li, H. Wang, Z. Wu, F. Dai, D. Cao, Vision-based detection and visualization of dynamic workspaces, *Autom. Constr.* 104 (2019) 1–13, <https://doi.org/10.1016/j.autcon.2019.04.001>.
 - [27] Y. Pi, N.D. Nath, A.H. Behzadan, Convolutional neural networks for object detection in aerial imagery for disaster response and recovery, *Adv. Eng. Inform.* 43 (2020) 101009, <https://doi.org/10.1016/j.aei.2019.101009>.
 - [28] H. Kim, S. Bang, H. Jeong, Y. Ham, H. Kim, Analyzing context and productivity of tunnel earthmoving processes using imaging and simulation, *Autom. Constr.* 92 (2018) 188–198, <https://doi.org/10.1016/j.autcon.2018.04.002>.
 - [29] H. Kim, Y. Ham, W. Kim, S. Park, H. Kim, Vision-based nonintrusive context documentation for earthmoving productivity simulation, *Autom. Constr.* 102 (2019) 135–147, <https://doi.org/10.1016/j.autcon.2019.02.006>.
 - [30] X. Luo, H. Li, D. Cao, F. Dai, J. Seo, S. Lee, Recognizing diverse construction activities in site images via relevance networks of construction-related objects detected by convolutional neural networks, *J. Comput. Civ. Eng.* 32 (2018), 04018012, [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000756](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000756).
 - [31] X. Luo, H. Li, X. Yang, Y. Yu, D. Cao, Capturing and understanding workers' activities in far-field surveillance videos with deep action recognition and bayesian nonparametric learning, *Comput.-Aided Civ. Infrastruct. Eng.* 34 (2019) 333–351, <https://doi.org/10.1111/mice.12419>.
 - [32] J. Kim, S. Chi, Action recognition of earthmoving excavators based on sequential pattern analysis of visual features and operation cycles, *Autom. Constr.* 104 (2019) 255–264, <https://doi.org/10.1016/j.autcon.2019.03.025>.
 - [33] C. Chen, Z. Zhu, A. Hammad, Automated excavators activity recognition and productivity analysis from construction site surveillance videos, *Autom. Constr.* 110 (2020) 103045, <https://doi.org/10.1016/j.autcon.2019.103045>.
 - [34] J. Cai, Y. Zhang, H. Cai, Two-step long short-term memory method for identifying construction activities through positional and attentional cues, *Autom. Constr.* 106 (2019) 102886, <https://doi.org/10.1016/j.autcon.2019.102886>.
 - [35] S. Bang, H. Kim, Context-based information generation for managing UAV-acquired data using image captioning, *Autom. Constr.* 112 (2020) 103116, <https://doi.org/10.1016/j.autcon.2020.103116>.
 - [36] D. Kim, M. Liu, S. Lee, V.R. Kamat, Remote proximity monitoring between mobile construction resources using camera-mounted UAVs, *Autom. Constr.* 99 (2019) 168–182, <https://doi.org/10.1016/j.autcon.2018.12.014>.
 - [37] H. Son, H. Seong, H. Choi, C. Kim, Real-time vision-based warning system for prevention of collisions between workers and heavy equipment, *J. Comput. Civ. Eng.* 33 (2019), 04019029, [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000845](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000845).
 - [38] J. Wu, N. Cai, W. Chen, H. Wang, G. Wang, Automatic detection of hardhats worn by construction personnel: a deep learning approach and benchmark dataset, *Autom. Constr.* 106 (2019) 102894, <https://doi.org/10.1016/j.autcon.2019.102894>.
 - [39] Q. Fang, H. Li, X. Luo, L. Ding, H. Luo, C. Li, Computer vision aided inspection on falling prevention measures for steepclimbs in an aerial environment, *Autom. Constr.* 93 (2018) 148–164, <https://doi.org/10.1016/j.autcon.2018.05.022>.
 - [40] Q. Fang, H. Li, X. Luo, L. Ding, T.M. Rose, W. An, Y. Yu, A deep learning-based method for detecting non-certified work on construction sites, *Adv. Eng. Inform.* 35 (2018) 56–68, <https://doi.org/10.1016/j.aei.2018.01.001>.
 - [41] L. Ding, W. Fang, H. Luo, P.E.D. Love, B. Zhong, X. Ouyang, A deep hybrid learning model to detect unsafe behavior: integrating convolution neural networks and long short-term memory, *Autom. Constr.* 86 (2018) 118–124, <https://doi.org/10.1016/j.autcon.2017.11.002>.
 - [42] M.M. Soltani, Z. Zhu, A. Hammad, Automated annotation for visual recognition of construction resources using synthetic images, *Autom. Constr.* 62 (2016) 14–23, <https://doi.org/10.1016/j.autcon.2015.10.002>.
 - [43] A. Braun, A. Borrmann, Combining inverse photogrammetry and BIM for automated labeling of construction site images for machine learning, *Autom. Constr.* 106 (2019) 102879, <https://doi.org/10.1016/j.autcon.2019.102879>.
 - [44] B. Kang, Z. Liu, X. Wang, F. Yu, J. Feng, T. Darrell, Few-shot object detection via feature reweighting, in: Proceedings of 2019 IEEE International Conference on Computer Vision 2019, IEEE, Seoul, Republic of Korea, 2019, pp. 8420–8429, https://openaccess.thecvf.com/content/ICCV_2019/papers/Kang_Few-Shot_Object_Detection_via_Feature_Reweighting_ICCV_2019_paper.pdf. (Accessed 16 November 2020).
 - [45] P. Brazdil, C.G. Carrier, C. Soares, R. Vilalta, *Metalearning*, Springer, 2009. <https://www.springer.com/gp/book/9783540732624> (accessed Nov 16, 2020) (ISBN: 978-3-540-73263-1).
 - [46] A. Rivolli, L.P.F. Garcia, C. Soares, J. Vanschoren, A.C.P.L.F. de Carvalho, Characterizing Classification Datasets: A Study of Meta-features for Meta-learning, <http://arxiv.org/abs/1808.10406>, 2018 (accessed Nov 11, 2020).
 - [47] F. Pinto, C. Soares, J. Mendes-Moreira, Towards Automatic Generation of Metafeatures, in: Proceedings of 20th Pacific-Asia Conference on Knowledge Discovery and Data Mining 2016, Springer, Auckland, New Zealand, 2016, pp. 215–226, https://doi.org/10.1007/978-3-319-31753-3_18.
 - [48] J. Redmon, A. Farhadi, YOLO9000: Better, Faster, Stronger, in: Proceedings of the 30th IEEE Conference on Computer Vision and Pattern Recognition, IEEE, Los Alamitos, U.S., 2017, pp. 6517–6525, <https://doi.org/10.1109/CVPR.2017.690>.
 - [49] S. Bang, F. Baek, S. Park, W. Kim, H. Kim, Image augmentation to improve construction resource detection using generative adversarial networks, cut-and-paste, and image transformation techniques, *Autom. Constr.* 115 (2020) 103198, <https://doi.org/10.1016/j.autcon.2020.103198>.
 - [50] O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein, A.C. Berg, L. Fei-Fei, ImageNet large scale visual recognition challenge, *Int. J. Comput. Vis.* 115 (2015) 211–252, <https://doi.org/10.1007/s11263-015-0816-y>.
 - [51] Y.X. Wang, D. Ramanan, M. Hebert, Meta-learning to detect rare objects, in: Proceedings of the IEEE International Conference on Computer Vision, IEEE, Seoul,

- Republic of Korea, 2019, pp. 9924–9933, <https://doi.org/10.1109/ICCV.2019.01002>.
- [52] X. Yan, Z. Chen, A. Xu, X. Wang, X. Liang, L. Lin, Meta R-CNN: Towards general solver for instance-level low-shot learning, in: Proceedings of 2019 IEEE International Conference on Computer Vision, IEEE, Seoul, Republic of Korea, 2019, pp. 9576–9585, <https://doi.org/10.1109/ICCV.2019.00967>.
- [53] M. Everingham, L. Van Gool, C.K.I. Williams, J. Winn, A. Zisserman, The pascal visual object classes (VOC) challenge, *Int. J. Comput. Vis.* 88 (2010) 303–338, <https://doi.org/10.1007/s11263-009-0275-4>.
- [54] J. Kim, S. Chi, Multi-camera vision-based productivity monitoring of earthmoving operations, *Autom. Constr.* 112 (2020) 103121, <https://doi.org/10.1016/j.autcon.2020.103121>.
- [55] J. Kim, S. Chi, M. Choi, Sequential pattern learning of visual features and operation cycles for vision-based action recognition of earthmoving excavators, in: Computing in Civil Engineering 2019: Data, Sensing, and Analytics, American Society of Civil Engineers, Atlanta, U.S., 2019, pp. 298–304, <https://doi.org/10.1061/9780784479247.083>.
- [56] J. Kim, S. Chi, B.-G. Hwang, Vision-based activity analysis framework considering interactive operation of construction equipment, in: ASCE International Workshop on Computing in Civil Engineering 2017, American Society of Civil Engineers, Seattle, U.S., 2017, pp. 162–170, <https://doi.org/10.1061/9780784480830.021>.
- [57] J. Kim, S. Chi, J. Seo, Interaction analysis for vision-based activity identification of earthmoving excavators and dump trucks, *Autom. Constr.* 87 (2018) 297–308, <https://doi.org/10.1016/J.AUTCON.2017.12.016>.
- [58] L. van der Maaten, G. Hinton, Visualizing Data using t-SNE, *J. Mach. Learn. Res.* 9 (2008) 2579–2605, doi:10.1.1.457.7213.
- [59] L. Yang, S. Hanneke, J. Carbonell, A theory of transfer learning with applications to active learning, *Mach. Learn.* 90 (2013) 161–189, <https://doi.org/10.1007/s10994-012-5310-y>.
- [60] J. Kim, J. Hwang, S. Chi, J. Seo, Towards database-free vision-based monitoring on construction sites: a deep active learning approach, *Autom. Constr.* 120 (2020) 103376, <https://doi.org/10.1016/j.autcon.2020.103376>.
- [61] K. Min, M. Jung, J. Kim, S. Chi, Sound event recognition-based classification model for automated emergency detection in indoor environment, in: Advances in Informatics and Computing in Civil and Construction Engineering, Springer, Chicago, U.S., 2018, pp. 529–535, https://doi.org/10.1007/978-3-030-00220-6_63.
- [62] H. Kim, C.R. Ahn, D. Engelhaupt, S. Lee, Application of dynamic time warping to the recognition of mixed equipment activities in cycle time measurement, *Autom. Constr.* 87 (2018) 225–234, <https://doi.org/10.1016/J.AUTCON.2017.12.014>.
- [63] J. Kim, K. Min, M. Jung, S. Chi, Occupant behavior monitoring and emergency event detection in single-person households using deep learning-based sound recognition, *Build. Environ.* 181 (2020) 107092, <https://doi.org/10.1016/j.buildenv.2020.107092>.
- [64] T. Kim, S. Chi, Accident case retrieval and analyses: using natural language processing in the construction industry, *J. Constr. Eng. Manag.* 145 (2019), 04019004, [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0001625](https://doi.org/10.1061/(ASCE)CO.1943-7862.0001625).
- [65] S. Moon, G. Lee, S. Chi, H. Oh, Automated construction specification review with named entity recognition using natural language processing, *J. Constr. Eng. Manag.* 147 (2021), 04020147, [https://doi.org/10.1061/\(asce\)co.1943-7862.0001953](https://doi.org/10.1061/(asce)co.1943-7862.0001953).
- [66] S. Moon, S. Chung, S. Chi, Bridge damage recognition from inspection reports using NER based on recurrent neural network with active learning, *J. Perform. Constr. Facil.* 34 (2020), 04020119, [https://doi.org/10.1061/\(asce\)cf.1943-5509.0001530](https://doi.org/10.1061/(asce)cf.1943-5509.0001530).